

Diabetic Retinopathy Detection using ML

A PROJECT REPORT

Major Project – II (01CE0807)

Submitted by

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Major Project-II (01CE0807)

Department of Computer Engineering

Faculty of Engineering & Technology

Marwadi University

A.Y. 2025-26

CERTIFICATE

This is to certify that the project report submitted along with the project entitled Diabetic Retinopathy Detection using ML has been carried out by Sanjana Ghinaiya (92200103150) under my guidance in partial fulfilment for the degree of Bachelor of Technology in Computer Engineering, 8th Semester of Marwadi University, Rajkot during the academic year 2025-26.

Dr.Anjali Diwan

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DECLARATION

We hereby declare that the **Major Project-II (01CE0807)** report submitted along with the Project entitled Diabetic Retinopathy Detection using ML submitted in partial fulfilment for the degree of Bachelor of Technology in Computer Engineering to Marwadi University, Rajkot, is a bonafide record of original project work carried out by me at Marwadi University under the supervision of Dr.Anjali Diwan and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

Name of the Student

Sign of Student

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I would like to express my sincere gratitude to Marwadi University, Faculty of Engineering & Technology, Department of Computer Engineering, for providing me with the opportunity to undertake this Major Project-II. The university's academic environment, research-oriented culture, and well-equipped infrastructure significantly supported the successful execution of this project. I am thankful to the institution for encouraging practical learning and technical innovation.

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Abstract

Diabetic Retinopathy is a severe eye disorder caused by prolonged diabetes and is one of the leading reasons for vision impairment and blindness across the world. Early detection plays a vital role in preventing permanent vision loss; however, traditional diagnosis methods rely heavily on manual examination of retinal fundus images by trained ophthalmologists, which is time-consuming and resource-intensive. Therefore, there is a strong need for an automated and reliable system that can assist medical professionals in early and accurate detection of diabetic retinopathy.

This project presents a deep learning–based approach for automated detection and classification of diabetic retinopathy using retinal fundus images. Several convolutional neural network architectures, including ResNet50, DenseNet121, EfficientNet-B0, and EfficientNet-B4, are implemented using transfer learning to extract meaningful features from retinal images. Image preprocessing techniques such as resizing, normalization, and data augmentation are applied to improve model robustness and generalization. The performance of the models is evaluated using metrics such as accuracy, quadratic weighted kappa, confusion matrix, and training and validation accuracy and loss curves.

The experimental results indicate that deep learning models are highly effective in identifying different stages of diabetic retinopathy, with EfficientNet-based architectures achieving superior performance compared to other models. The proposed system demonstrates reliable classification capability and strong agreement with actual labels, making it suitable for automated screening applications. This approach has the potential to reduce the workload of ophthalmologists and enable early diagnosis, thereby improving patient outcomes and supporting large-scale diabetic retinopathy screening programs.

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Abbreviations

DR	Diabetic Retinopathy
CNN	Convolutional Neural Network
DL	Deep Learning
ML	Machine Learning
ResNet	Residual Neural Network
DenseNet	Densely Connected Convolutional Network
EfficientNet	Efficient Convolutional Neural Network
GPU	Graphics Processing Unit
QWK	Quadratic Weighted Kappa
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
ReLU	Rectified Linear Unit
Softmax	Softmax Activation Function
SGD	Stochastic Gradient Descent
Adam	Adaptive Moment Estimation

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CHAPTER 1

INTRODUCTION

1.1 PROJECT SUMMARY

Diabetic Retinopathy is a severe eye disease caused by prolonged diabetes and is one of the leading causes of vision impairment and blindness worldwide. Early detection of diabetic retinopathy is crucial for preventing permanent vision loss; however, traditional diagnosis relies on manual examination of retinal fundus images by trained ophthalmologists. This process is time-consuming, subjective, and difficult to scale for large populations. With the increasing number of diabetic patients, there is a strong need for an automated and efficient diagnostic system to support early screening and timely treatment.

This project presents a deep learning-based system for the automated detection and classification of diabetic retinopathy using retinal fundus images. The proposed system utilizes convolutional neural network architectures to automatically learn complex visual patterns associated with different stages of diabetic retinopathy. To enhance performance and reduce training time, transfer learning is applied using pre-trained models such as ResNet50, DenseNet121, EfficientNet-B0, and EfficientNet-B4. Image preprocessing techniques including resizing, normalization, and data augmentation are performed to improve model robustness and generalization.

The system is trained and evaluated using labeled retinal image datasets. Multiple performance metrics such as accuracy, quadratic weighted kappa, confusion matrix, and training-validation loss and accuracy curves are used to assess model effectiveness. Experimental results demonstrate that deep learning models, particularly EfficientNet-based architectures, provide high accuracy and reliable classification performance. The automated system significantly reduces human effort and enables consistent and objective analysis of retinal images.

In conclusion, the proposed diabetic retinopathy detection system offers an efficient, scalable, and accurate solution for automated retinal screening. By assisting ophthalmologists in early diagnosis, the system has the potential to improve clinical decision-making and reduce the burden on healthcare systems. With further optimization and clinical validation, this approach can be integrated into real-world medical screening applications for large-scale diabetic retinopathy detection.

1.2 PROJECT PURPOSE

The primary purpose of this project is to develop an automated and intelligent system for the detection and classification of diabetic retinopathy using deep learning techniques. Diabetic retinopathy is a serious eye disease caused by prolonged diabetes and is one of the leading contributors to vision impairment and blindness worldwide. Early identification of the disease plays a critical role in preventing irreversible vision loss; however, conventional diagnostic methods rely heavily on manual examination of retinal fundus images by experienced ophthalmologists. Such methods are time-consuming, prone to subjectivity, and difficult to scale for the growing diabetic population. This project aims to address these challenges by introducing a computer-aided diagnostic approach that improves efficiency and consistency in screening.

Another important purpose of this project is to explore the applicability of advanced convolutional neural network architectures in medical image analysis. The project focuses on leveraging transfer learning with pre-trained deep learning models to automatically extract meaningful features from retinal images without the need for handcrafted feature engineering. By implementing multiple CNN-based models and analyzing their performance, the project seeks to identify an effective architecture for diabetic retinopathy detection. Image preprocessing and data augmentation techniques are also incorporated to enhance model generalization and robustness against variations in image quality.

Furthermore, the project aims to support large-scale diabetic retinopathy screening programs by providing a reliable and scalable diagnostic solution. The proposed system is designed to assist healthcare professionals by delivering accurate and objective predictions, thereby reducing diagnostic workload and enabling timely clinical decision-making. By improving early detection and classification accuracy, the system contributes toward better patient management and improved healthcare outcomes. Ultimately, the purpose of this project is to demonstrate how deep learning technologies can be effectively applied in medical imaging to enhance diagnostic capabilities and support preventive healthcare initiatives.

Key Purpose of the Project:

- To develop an automated deep learning-based system for early detection and classification of diabetic retinopathy using retinal fundus images.
- To reduce reliance on manual diagnosis by ophthalmologists and improve the accuracy, consistency, and speed of retinal disease screening.
- To utilize convolutional neural networks and transfer learning techniques for effective feature extraction and classification of diabetic retinopathy stages.

- To provide a scalable and reliable solution that supports large-scale screening and assists healthcare professionals in timely clinical decision-making.

1.3 OBJECTIVE

- To design and implement an automated deep learning–based system for the detection and classification of diabetic retinopathy using retinal fundus images.
- To study and analyze various convolutional neural network architectures suitable for medical image classification.
- To apply transfer learning techniques using pre-trained deep learning models to enhance performance and reduce computational cost.
- To perform image preprocessing and data augmentation to improve data quality and model generalization.
- To train deep learning models on labeled retinal image datasets and optimize hyperparameters for improved accuracy.
- To evaluate model performance using appropriate metrics such as accuracy, quadratic weighted kappa, confusion matrix, and loss curves.
- To compare the performance of multiple deep learning models and identify the most effective architecture for diabetic retinopathy detection.
- To minimize false diagnosis by improving classification consistency and reliability.
- To reduce the time and effort required for manual screening of retinal images by ophthalmologists.
- To develop a scalable system that can be integrated into large-scale diabetic retinopathy screening programs.
- To assist healthcare professionals by providing objective and interpretable diagnostic results.
- To demonstrate the practical application of deep learning techniques in medical image analysis.

1.4 SCOPE

The scope of this project is focused on the development of an automated system for the detection and classification of diabetic retinopathy using deep learning techniques. The project involves the use of retinal fundus images as input data and applies convolutional neural network–based models to identify different stages of diabetic retinopathy. The system is designed to assist in early diagnosis and support ophthalmologists by providing accurate and consistent predictions.

This project includes image preprocessing steps such as resizing, normalization, and data augmentation to improve model performance and generalization. Transfer learning techniques using pre-trained deep learning architectures are employed to enhance feature extraction and reduce training complexity. The scope also covers training, validation, and performance evaluation of multiple deep learning models using appropriate metrics such as accuracy and quadratic weighted kappa.

The project is limited to the analysis of retinal fundus images and does not include real-time image acquisition or direct clinical deployment. The system is intended as a decision-support tool rather than a replacement for medical professionals. Future extensions such as real-world clinical validation, integration with hospital systems, and deployment as a web or mobile application are beyond the current scope of this project.

1.5 TECHNOLOGY AND LITERATURE REVIEW

Recent advancements in deep learning have significantly improved automated medical image analysis, particularly in the detection of diabetic retinopathy from retinal fundus images. Diabetic retinopathy is a leading cause of vision impairment, and early diagnosis is essential to prevent irreversible damage. Traditional diagnosis methods rely on manual inspection by ophthalmologists or classical image processing techniques, which are time-consuming and often limited in accuracy. To overcome these challenges, researchers have increasingly adopted convolutional neural networks due to their ability to automatically learn complex visual features from images.

Several studies have demonstrated that deep learning-based approaches outperform traditional machine learning methods in diabetic retinopathy classification. Pre-trained CNN architectures such as ResNet, DenseNet, and EfficientNet have been widely used through transfer learning to improve performance while reducing training time and computational cost. These models, trained on large-scale image datasets, are fine-tuned using retinal images to effectively capture disease-specific features such as lesions and vascular abnormalities. Literature also highlights the importance of image preprocessing techniques, including resizing, normalization, and data augmentation, to enhance model robustness and generalization.

In this project, deep learning frameworks such as TensorFlow and Keras are used for model development and training, with GPU-enabled environments accelerating the learning process. Performance evaluation metrics such as accuracy and quadratic weighted kappa are commonly employed in existing studies and are also adopted in this work to ensure reliable assessment. By integrating proven deep learning technologies and building upon findings from existing research, this project aims to develop an effective and scalable system for automated diabetic retinopathy detection.

1.6 PROJECT SCHEDULING

Project scheduling is an important aspect of effective project management, as it helps in planning, organizing, and tracking project activities within a defined time frame. The scheduling of this project was carried out by dividing the entire work into multiple phases and allocating a specific duration to each phase. This systematic approach ensured timely completion of tasks and smooth progress throughout the project lifecycle.

The project began with problem identification and literature review to understand diabetic retinopathy, existing diagnostic methods, and the application of deep learning in medical imaging. This was followed by dataset collection and preprocessing, which included image resizing, normalization, and data augmentation. The next phase involved the implementation

of deep learning models using transfer learning techniques and training them on the prepared dataset. Model evaluation and performance analysis were carried out using appropriate metrics to assess accuracy and reliability.

In the final phase, results were analyzed, compared, and documented, followed by preparation of the project report and presentation. A Gantt chart was used to represent the project schedule visually, highlighting task durations and dependencies. The planned schedule helped in efficient time management and ensured that the project milestones were achieved within the stipulated academic timeline.



Fig 1.1 Project Complete Work Schedule

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

In the existing system, diabetic retinopathy is primarily diagnosed through manual examination of retinal fundus images by trained ophthalmologists. The diagnosis process involves visual inspection to identify abnormalities such as microaneurysms, hemorrhages, and exudates. While this approach is clinically effective, it requires significant expertise, is time-consuming, and is not scalable for screening a large number of diabetic patients. Early-stage diabetic retinopathy can also be difficult to detect through manual observation, leading to delayed diagnosis in some cases.

Some automated systems have been developed using traditional image processing and machine learning techniques. These systems rely on handcrafted feature extraction methods to detect retinal abnormalities, followed by conventional classifiers. However, such approaches are highly dependent on image quality and feature design, which limits their accuracy and generalization capability. Additionally, traditional methods struggle to capture complex patterns present in retinal images, making them less reliable for multi-stage classification of diabetic retinopathy.

Overall, the existing systems face challenges related to efficiency, consistency, and scalability. The reliance on manual diagnosis and conventional techniques increases the risk of human error and makes it difficult to implement large-scale screening programs. These limitations highlight the need for an automated and intelligent diagnostic system that can provide accurate and consistent results while reducing the workload on medical professionals.

2.2 LIMITATIONS OF EXISTING SYSTEM

- **Manual Diagnosis Dependency**

The existing system largely depends on manual examination of fundus images by ophthalmologists, which is time-consuming and prone to human error, especially in early-stage detection.

- **Limited Accuracy in Early Stages**

Traditional and conventional machine-learning approaches struggle to accurately detect mild and moderate stages of diabetic retinopathy due to subtle retinal features.

- **High Inter-Observer Variability**

Diagnosis results may vary between different experts, leading to inconsistency and reduced reliability in disease grading.

- **Scalability Issues**

Manual screening methods are not scalable for large populations, particularly in regions with limited access to trained medical professionals.

- **Sensitivity to Image Quality**

Existing systems often fail when fundus images have poor illumination, noise, blur, or artifacts, which affects overall performance.

- **Limited Feature Extraction Capability**

Traditional methods rely on handcrafted features, which may not capture complex patterns such as microaneurysms and exudates effectively.

- **High Cost and Time Consumption**

Regular screening using existing systems requires specialized equipment and expert analysis, increasing operational cost and diagnosis time.

- **Delayed Treatment Initiation**

Due to slow screening and diagnosis processes, early intervention is often delayed, increasing the risk of vision loss.

- **Lack of Automation**

The existing system lacks full automation, requiring continuous human involvement for screening, grading, and validation, which reduces efficiency.

- **Poor Generalization Across Datasets**

Models trained using traditional techniques often perform poorly on images from different cameras, datasets, or clinical environments.

- **Imbalanced Dataset Handling**

Existing systems struggle with class imbalance, especially when severe cases are fewer than normal or mild cases, leading to biased predictions.

- **Inconsistent Performance Across Severity Levels**

The diagnostic accuracy varies significantly across different stages of diabetic retinopathy, with lower performance in borderline and early stages.

- **Limited Robustness to Real-World Conditions**

Variations in lighting conditions, camera resolution, and patient movement adversely affect the performance of existing systems.

- **Absence of Real-Time Analysis**

Existing systems do not support real-time screening and analysis, making them unsuitable for immediate clinical decision-making.

- **Difficulty in Detecting Multiple Lesions Simultaneously**

Handcrafted-feature-based methods struggle to detect multiple retinal lesions appearing together in a single fundus

2.3 PROPOSED SYSTEM

The proposed system aims to overcome the limitations of the existing manual and traditional automated methods by introducing a deep learning-based approach for the detection and classification of diabetic retinopathy. The system is designed to automatically analyze retinal fundus images and accurately identify different stages of diabetic retinopathy with minimal human intervention. By leveraging convolutional neural networks, the proposed system is capable of learning complex visual patterns present in retinal images that are difficult to detect through manual examination.

In the proposed approach, retinal fundus images are first preprocessed to ensure consistency and improve image quality. Preprocessing steps include image resizing, normalization, and data augmentation techniques such as rotation and flipping to enhance model generalization. These processed images are then fed into deep learning models using transfer learning. Pre-trained architectures such as ResNet50, DenseNet121, and EfficientNet variants are fine-tuned for the diabetic retinopathy classification task, enabling effective feature extraction and improved accuracy while reducing training time.

The system classifies retinal images into predefined diabetic retinopathy severity levels and provides reliable predictions based on learned features. Performance evaluation is carried out using appropriate metrics such as accuracy and quadratic weighted kappa to ensure robust and consistent results. The proposed system is scalable and suitable for large-scale screening applications, making it an effective decision-support tool for ophthalmologists. By automating the screening process, the system reduces diagnostic time, minimizes human error, and supports early detection of diabetic retinopathy, thereby improving patient care and clinical outcomes.

2.4 FEATURES OF PROPOSED SYSTEM

- **Automated Detection:** The proposed system automatically detects and classifies diabetic retinopathy from retinal fundus images, reducing the need for manual examination and minimizing human intervention.
- **High Accuracy:** By utilizing deep learning and convolutional neural network architectures, the system achieves high classification accuracy in identifying different stages of diabetic retinopathy.
- **Multi-Stage Classification:** The system is capable of classifying retinal images into multiple diabetic retinopathy severity levels, enabling detailed and reliable diagnosis.
- **Transfer Learning Approach:** Pre-trained deep learning models are used to enhance feature extraction, reduce training time, and improve overall performance.
- **Robust Preprocessing:** Image preprocessing and data augmentation techniques improve model generalization and handle variations in image quality.

- **Scalability:** The proposed system can be easily scaled for large-scale screening of diabetic patients, making it suitable for use in mass screening programs.
- **Consistency and Reliability:** The automated approach provides consistent diagnostic results, reducing variability caused by human interpretation.
- **Decision Support:** The system acts as a clinical decision-support tool by assisting ophthalmologists in early detection and timely treatment planning.
- **Early Disease Detection:** Enables identification of diabetic retinopathy at early stages, supporting timely intervention and treatment.
- **Consistent Predictions:** Produces uniform diagnostic results for similar inputs, reducing variability caused by subjective human judgment.
- **Reduced Diagnostic Time:** Significantly decreases the time required to analyze retinal images compared to manual screening.
- **Handling Large Datasets:** Capable of processing large volumes of retinal images efficiently, making it suitable for population-level screening.
- **Model Generalization:** Uses data augmentation and transfer learning to improve performance across diverse retinal image datasets.
- **Reduced Clinical Workload:** Helps reduce the workload of ophthalmologists by automating routine screening tasks.
- **Extensible Framework:** The system architecture can be extended to detect other retinal diseases with minimal modifications.
- **Research and Educational Use:** Can be used as a research tool for studying deep learning applications in medical image analysis.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM DESIGN AND METHODOLOGY

The proposed system is designed to provide an automated and reliable solution for detecting and classifying diabetic retinopathy using deep learning techniques. The system follows a structured and modular design approach to ensure accuracy, scalability, and ease of understanding. The methodology integrates image preprocessing, deep learning-based feature extraction, and classification using transfer learning models. The overall design aims to reduce manual effort and assist healthcare professionals in early diagnosis of diabetic retinopathy.

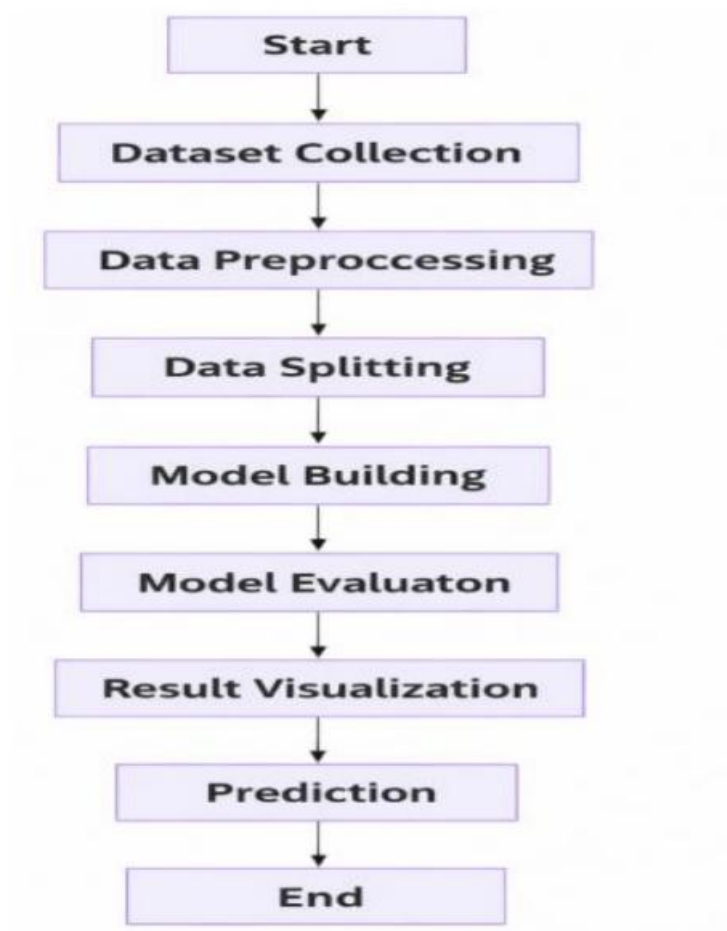


Fig 3.1: Workflow

3.1.1 System Overview

The system accepts retinal fundus images as input and processes them through a deep learning pipeline to determine the severity of diabetic retinopathy. The workflow includes image acquisition from the dataset, preprocessing and augmentation, feature extraction using convolutional neural networks, and classification. Transfer learning is employed to utilize pre-trained deep learning models, which improves accuracy while reducing training time. The final output of the system is a predicted diabetic retinopathy class along with confidence information.

3.1.2 Dataset Utilization

The system uses the APTOS 2019 Blindness Detection dataset, which contains labeled retinal fundus images representing different stages of diabetic retinopathy. The dataset includes images captured under varying imaging conditions, such as differences in illumination and resolution. These variations make the dataset suitable for training robust deep learning models. The dataset is divided into training and validation subsets to support supervised learning and reliable performance evaluation.

3.1.3 Image Preprocessing and Data Augmentation

Image preprocessing is performed to standardize the input data and enhance image quality. Preprocessing steps include resizing images to a fixed resolution, normalizing pixel values, and reducing noise. To improve generalization and address class imbalance, data augmentation techniques such as rotation, flipping, zooming, and brightness adjustment are applied during training. These techniques increase dataset diversity and reduce overfitting.

3.1.4 Model Selection and Architecture Design

Convolutional neural network architectures are employed for feature extraction and classification due to their effectiveness in image-based tasks. Pre-trained models such as ResNet50, DenseNet121, EfficientNet-B0, and EfficientNet-B4 are selected for implementation. Transfer learning is applied by fine-tuning the upper layers of these models using the APTOS 2019 dataset, allowing the system to learn disease-specific features efficiently.

3.1.4.1 ResNet50

ResNet50 is a deep convolutional neural network architecture that belongs to the Residual Network family. It consists of 50 layers and introduces the concept of residual learning, where shortcut connections allow the network to skip one or more layers. These residual connections help overcome the vanishing gradient problem, enabling the training of deeper networks with improved stability. ResNet50 has been widely adopted in image classification tasks due to its ability to learn complex visual features effectively.

In this project, ResNet50 is used through transfer learning to extract meaningful features from retinal fundus images for diabetic retinopathy detection. The pre-trained weights allow the model to capture low-level and high-level features such as edges, textures, and lesion patterns present in retinal images. Its deep architecture makes it suitable for identifying subtle disease-related features, thereby improving classification accuracy.

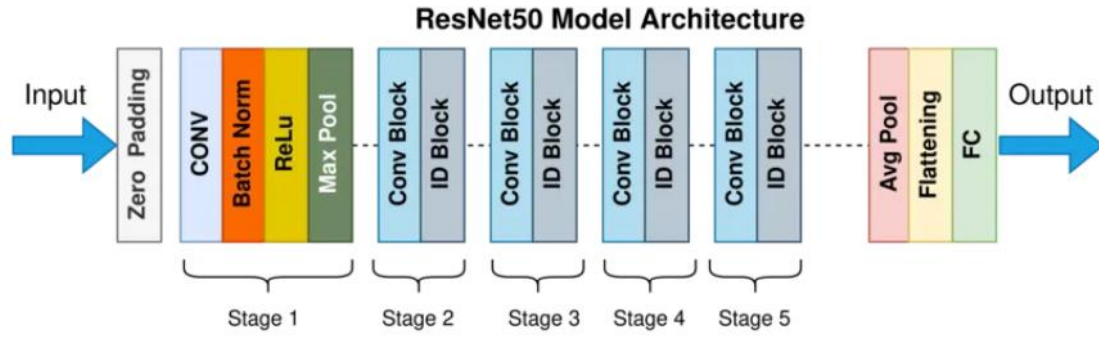


Fig 3.2: ResNet50 Architecture

3.1.4.2 DenseNet121

DenseNet121 is a convolutional neural network architecture characterized by dense connectivity between layers. In DenseNet, each layer receives feature maps from all preceding layers, which encourages feature reuse and improves information flow throughout the network. This dense connectivity significantly reduces the number of parameters while maintaining high performance, making DenseNet121 efficient and effective for image classification tasks.

For diabetic retinopathy detection, DenseNet121 enables the model to learn fine-grained retinal features by reusing previously learned representations. This property is particularly beneficial in medical image analysis, where capturing subtle variations is critical. By leveraging transfer learning, DenseNet121 enhances feature extraction while reducing overfitting, resulting in reliable classification performance.

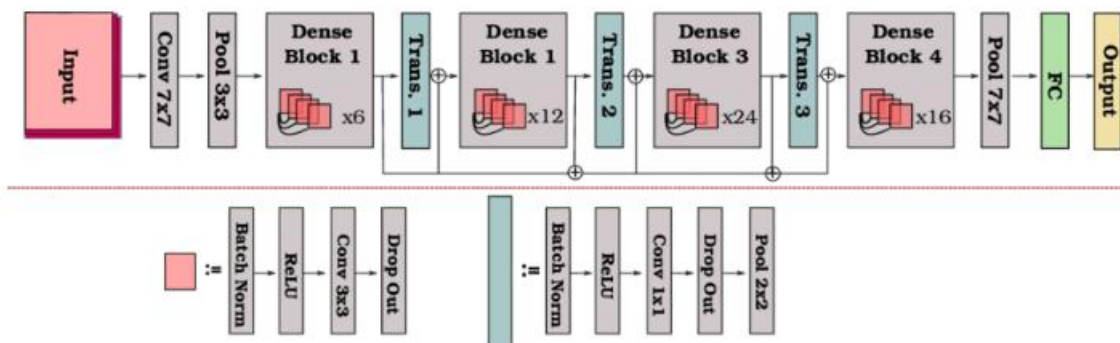


Fig 3.3: DenseNet121 Architecture

3.1.4.3 EfficientNet-B0

EfficientNet-B0 is a lightweight convolutional neural network architecture designed using a compound scaling method that balances network depth, width, and resolution. Unlike traditional scaling approaches, EfficientNet uniformly scales all dimensions of the network, resulting in improved accuracy with fewer parameters. EfficientNet-B0 serves as the baseline model in the EfficientNet family and is known for its computational efficiency.

In this project, EfficientNet-B0 is employed to achieve a balance between performance and computational cost. Its efficient design allows faster training and inference while maintaining competitive accuracy in diabetic retinopathy classification. The model is fine-tuned using transfer learning to adapt its learned features to retinal fundus images, making it suitable for large-scale screening applications.

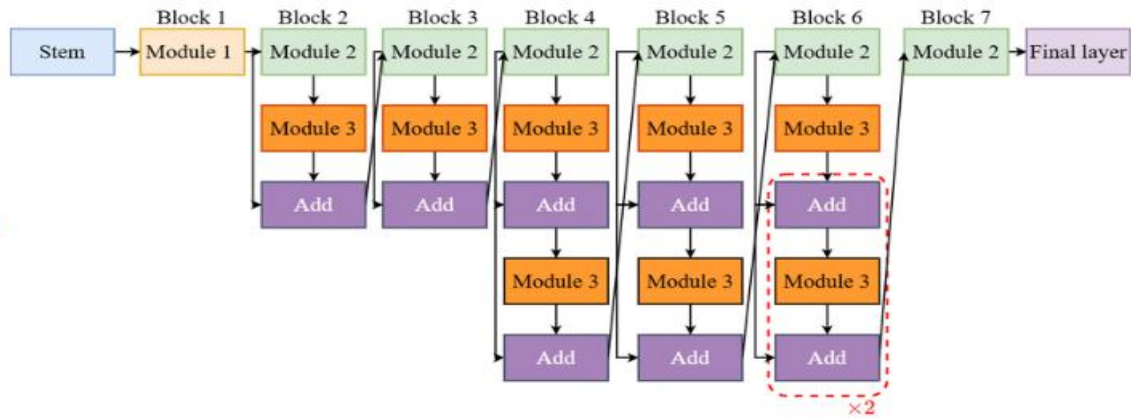


Fig 3.4: EfficientNetB0 Architecture

3.1.4.4 EfficientNet-B4

EfficientNet-B4 is a scaled-up version of the EfficientNet architecture, offering higher representational capacity compared to EfficientNet-B0. By increasing network depth, width, and input resolution in a balanced manner, EfficientNet-B4 achieves superior performance while maintaining computational efficiency. This architecture is particularly effective in complex image classification tasks that require detailed feature extraction.

In this project, EfficientNet-B4 is used to capture fine-grained retinal features associated with different stages of diabetic retinopathy. Its enhanced capacity enables improved discrimination between disease severity levels.

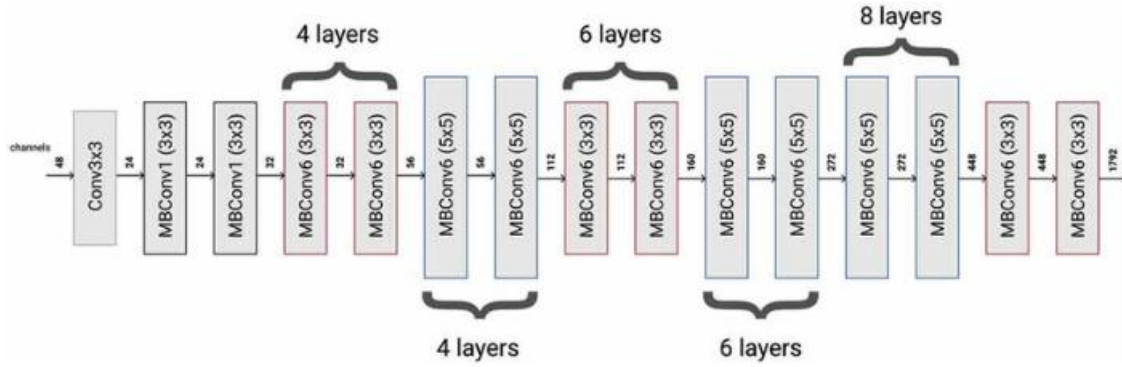


Fig 3.5.: EfficientNetB4 Architecture

3.1.5 Output Generation

After training, the models classify retinal fundus images into predefined diabetic retinopathy severity levels. The system outputs the predicted class label for each image. This output is designed to be clear and interpretable, making the system suitable as a decision-support tool for assisting ophthalmologists in diagnosis.

3.1.6 Performance Evaluation

Model performance is evaluated using metrics such as accuracy and quadratic weighted kappa. Confusion matrix analysis is performed to assess class-wise prediction performance. Training and validation accuracy and loss curves are also analyzed to study model convergence and stability. Comparative analysis of different models helps identify the most effective architecture for diabetic retinopathy detection.

3.2 SYSTEM ARCHITECTURE

The system architecture of the proposed diabetic retinopathy detection system is designed as a modular and sequential pipeline to ensure efficient processing and accurate classification of retinal fundus images. The architecture clearly defines the flow of data from image input to final prediction output. Each module performs a specific function, enabling clarity, scalability, and ease of maintenance. This structured design supports automated screening while reducing dependency on manual analysis.

The architecture begins with the input module, where retinal fundus images from the APTOS 2019 dataset are provided to the system. These images are captured under varying conditions and serve as raw input data. To handle variations in image quality, size, and illumination, the input images are passed to a preprocessing module that standardizes the data before further analysis.

The preprocessing module performs essential operations such as image resizing, normalization, and data augmentation to enhance image quality and consistency. The processed images are then forwarded to the deep learning module, which forms the core component of the system architecture. This module consists of convolutional neural network models implemented using transfer learning, enabling automatic extraction of meaningful features from retinal images.

Finally, the extracted features are passed to the classification module, which predicts the severity level of diabetic retinopathy for each input image. The output generated by the system represents the disease stage and serves as a reliable diagnostic aid. The modular architecture allows easy integration of additional models or datasets in the future, making the system adaptable and scalable for real-world screening applications.

3.3 DATASET AND DATA FLOW DESIGN

The dataset used in this project is the APTOS 2019 Blindness Detection dataset, which contains labeled retinal fundus images representing different stages of diabetic retinopathy. The dataset includes images captured under varying conditions of illumination, resolution, and retinal quality, reflecting real-world clinical scenarios. Each image is associated with a class label that indicates the severity level of diabetic retinopathy, making the dataset suitable for supervised deep learning-based classification.

- No DR: No sign of diabetic retinopathy.
- Mild: Early stage of the disease.
- Moderate: Intermediate stage.
- Severe: Advanced stage.
- Proliferative DR: Final stage, potential for severe vision loss.

The images exhibit high variability in resolution (ranging from approximately 321×321 to over 3000 pixels on the longer side), lighting conditions, focus quality, and artifacts, making the dataset realistic and challenging for deep learning models. The dataset includes only a labeled training set (accompanied by a CSV file mapping image IDs to diagnosis labels), while the test set remains unlabeled publicly for competition purposes. No separate validation set is provided, so participants commonly apply stratified k-fold cross-validation or manual splits.

The data flow begins with loading retinal fundus images from the dataset into the system. Since the images differ in size and quality, they are first passed through a preprocessing stage where resizing and normalization operations are applied. These steps ensure uniform input dimensions and consistent pixel intensity ranges, enabling effective learning by convolutional neural networks.

During the training phase, data augmentation techniques such as rotation, flipping, zooming, and brightness variation are applied to the training dataset. This increases dataset diversity and improves model generalization while reducing overfitting. The processed data is then divided into training and validation sets to facilitate reliable performance monitoring and unbiased evaluation during model training.

After training, the validation dataset is used to evaluate the trained models by comparing predicted class labels with actual ground truth labels. Performance metrics such as accuracy and quadratic weighted kappa are computed to assess classification effectiveness. This structured data flow ensures consistent data handling, efficient model training, and reliable evaluation of the proposed diabetic retinopathy detection system.

CHAPTER 4

IMPLEMENTATION AND TESTING

4.1 IMPLEMENTATION ENVIRONMENT

The implementation of the proposed diabetic retinopathy detection system was carried out using a deep learning–based software environment that supports efficient image processing and model training. The system was developed using the Python programming language due to its extensive support for machine learning and medical image analysis. Python provides flexibility, ease of implementation, and compatibility with multiple deep learning frameworks.

The deep learning models were implemented using popular libraries such as TensorFlow and Keras, which offer high-level APIs for designing, training, and evaluating convolutional neural networks. These libraries enable efficient use of transfer learning techniques and provide built-in functions for model optimization and performance evaluation. Additional libraries such as NumPy, Pandas, and OpenCV were used for numerical computation, data handling, and image preprocessing tasks.

The implementation was performed in a notebook-based development environment, allowing interactive experimentation and visualization of results. This environment facilitated easy debugging, model comparison, and analysis of training performance through graphical plots. The training process utilized GPU acceleration where available to reduce computational time and improve training efficiency.

Overall, the chosen implementation environment ensured efficient development, reliable execution, and reproducible results. The combination of Python, deep learning libraries, and hardware acceleration provided a robust platform for implementing and evaluating the proposed diabetic retinopathy detection system.

4.2 MODEL IMPLEMENTATION

The implementation of the proposed diabetic retinopathy detection system involves the development and fine-tuning of deep learning models using transfer learning techniques. Pre-trained convolutional neural network architectures such as ResNet50, DenseNet121, EfficientNet-B0, and EfficientNet-B4 were selected for implementation due to their proven performance in image classification tasks. These models were initialized with weights trained on large-scale image datasets, enabling efficient feature extraction from retinal fundus image.

The implementation begins by loading the selected pre-trained models without their final classification layers. Custom classification layers were then added to adapt the models for diabetic retinopathy severity classification. These layers include fully connected layers followed by appropriate activation functions to generate class probabilities. This approach allows the models to retain learned low-level visual features while learning disease-specific patterns from the APTOS 2019 dataset.

During implementation, the lower layers of the pre-trained models were initially frozen to preserve generic feature representations such as edges and textures. The upper layers were fine-tuned using the retinal fundus images to capture domain-specific features related to diabetic retinopathy. This selective training strategy reduces computational cost and prevents overfitting, especially when working with limited medical image datasets.

The training process was carried out using supervised learning, where labeled retinal images were provided as input to the models. An appropriate optimizer and loss function were selected to minimize classification error. Batch processing was used to efficiently train the models, and validation data was used to monitor training performance. Techniques such as early stopping were applied to prevent overfitting and ensure stable convergence.

Each model was trained independently using the same dataset and preprocessing pipeline to allow fair performance comparison. Model checkpoints were saved during training to retain the best-performing weights. After training, the models were evaluated on validation data to analyze accuracy, reliability, and generalization capability.

Overall, the model implementation process ensured efficient utilization of deep learning architectures and transfer learning techniques. The structured implementation enabled systematic comparison of multiple models and contributed to identifying the most effective architecture for automated diabetic retinopathy detection.

4.3 TRAINING STRATEGY

The training strategy for the proposed diabetic retinopathy detection system was designed to ensure effective learning, stable convergence, and robust generalization performance. The strategy follows a supervised learning approach, where labeled retinal fundus images from the APTOS 2019 dataset are used to train deep learning models. Special attention was given to handling class imbalance, preventing overfitting, and achieving optimal model performance.

Initially, transfer learning was employed by using pre-trained convolutional neural network models such as ResNet50, DenseNet121, EfficientNet-B0, and EfficientNet-B4. During the first phase of training, the lower layers of these models were frozen to preserve generic feature representations learned from large-scale image datasets. Only the newly added classification layers were trained, allowing the models to adapt to diabetic retinopathy-specific features while reducing computational complexity and training time.

After the initial convergence, a fine-tuning phase was carried out where selected higher layers of the pre-trained models were unfrozen. This allowed the models to learn more discriminative and domain-specific features from retinal fundus images. Fine-tuning was performed using a lower learning rate to avoid large weight updates and ensure stable learning. This two-stage training approach helped improve classification accuracy while minimizing the risk of overfitting.

To further enhance model generalization, data augmentation techniques such as rotation, flipping, zooming, and brightness adjustment were applied during training. Batch-based training was used to efficiently process large datasets, and validation data was utilized to monitor performance during each epoch. Early stopping was employed to halt training when validation performance stopped improving, thereby preventing overfitting and unnecessary computation.

Hyperparameters such as learning rate, batch size, and number of epochs were carefully selected based on experimental observations. Model checkpoints were saved to retain the best-performing model weights. The training progress was monitored using training and validation accuracy and loss curves, which provided insights into model convergence behavior.

Overall, the adopted training strategy ensured reliable learning, improved generalization, and fair comparison across multiple deep learning models. This systematic approach contributed significantly to achieving accurate and consistent diabetic retinopathy classification results.

4.4 TESTING AND EVALUATION

The testing and evaluation phase was conducted to assess the performance, reliability, and generalization capability of the trained deep learning models for diabetic retinopathy detection. After completion of the training process, the models were tested using validation data that was not used during training. This ensured an unbiased evaluation of the system's ability to classify unseen retinal fundus images accurately.

During testing, each trained model was used to predict the diabetic retinopathy severity level for the input retinal images. The predicted outputs were compared with the corresponding ground truth labels provided in the APTOS 2019 dataset. This comparison enabled objective measurement of classification performance and helped identify strengths and weaknesses of each model.

Performance evaluation was carried out using standard metrics such as accuracy and quadratic weighted kappa, which are widely used in medical image classification tasks. Confusion matrix analysis was also performed to analyze class-wise prediction behavior and to understand misclassification patterns. Additionally, training and validation accuracy and loss curves were examined to evaluate model convergence and stability.

A comparative analysis of all implemented models, including ResNet50, DenseNet121, EfficientNet-B0, and EfficientNet-B4, was conducted to determine the most effective architecture for diabetic retinopathy detection. This evaluation process ensured that the proposed system achieved reliable performance and provided meaningful insights into model behavior, supporting its suitability for automated diabetic retinopathy screening.

CHAPTER 5

RESULTS AND ANALYSIS

5.1 EXPERIMENTAL SETUP

The experimental setup was designed to evaluate the effectiveness and reliability of the proposed deep learning-based diabetic retinopathy detection system. All experiments were conducted using the APTOS 2019 Blindness Detection dataset, which contains labeled retinal fundus images representing different severity levels of diabetic retinopathy. The dataset was preprocessed and divided into training and validation sets to ensure fair and unbiased evaluation of the implemented models.

Before training, all retinal images were resized to a fixed input resolution compatible with the selected convolutional neural network architectures. Image normalization was applied to standardize pixel intensity values and improve numerical stability during training. Data augmentation techniques such as rotation, flipping, zooming, and brightness variation were applied to the training data to increase dataset diversity and reduce overfitting. The same preprocessing pipeline was consistently applied across all models to ensure a fair comparison.

Multiple deep learning models, including ResNet50, DenseNet121, EfficientNet-B0, and EfficientNet-B4, were implemented using transfer learning. Each model was initialized with pre-trained weights, and the final classification layers were customized to match the number of diabetic retinopathy classes. The models were trained independently using identical training conditions, including the same dataset split, preprocessing steps, and evaluation metrics, to ensure consistency in experimental results.

The training process was carried out using batch-based supervised learning. Model performance was monitored using validation data during training to track convergence and detect overfitting. Early stopping and model checkpointing techniques were employed to retain the best-performing model weights. After training, the validation dataset was used for testing, and predictions were generated for performance evaluation.

Overall, the experimental setup ensured a controlled and systematic evaluation of all implemented models. By maintaining consistent conditions across experiments, the setup enabled accurate comparison of model performance and provided reliable insights into the effectiveness of deep learning architectures for automated diabetic retinopathy detection.

5.2 PERFORMANCE METRICS

Performance metrics are used to quantitatively evaluate the effectiveness of the proposed deep learning models in detecting and classifying diabetic retinopathy. In this project, appropriate evaluation metrics were selected to measure both overall classification accuracy and agreement between predicted and actual severity levels. These metrics provide insight into model reliability, consistency, and practical usefulness in medical image analysis.

Accuracy is used as a primary metric to measure the proportion of correctly classified retinal images among the total number of samples. It provides an overall indication of how well the model performs across all classes. However, accuracy alone may not fully reflect model performance in datasets with class imbalance, which is common in medical imaging tasks such as diabetic retinopathy detection.

To address this limitation, Quadratic Weighted Kappa is used as an additional evaluation metric. It measures the level of agreement between predicted labels and ground truth labels while penalizing larger misclassification errors more heavily. This metric is particularly suitable for diabetic retinopathy classification, as it considers the ordered nature of disease severity levels and provides a more meaningful performance assessment.

In addition to accuracy, a confusion matrix is used to analyze class-wise performance of the models. The confusion matrix helps identify correctly classified instances as well as misclassification patterns between different diabetic retinopathy stages. Furthermore, training and validation accuracy and loss curves are analyzed to study model convergence, learning behavior, and potential overfitting during training.

Overall, the combination of these performance metrics ensures a comprehensive evaluation of the proposed system. By using multiple metrics, the evaluation process provides reliable insights into both overall and class-specific model performance, supporting objective comparison of different deep learning architectures.

5.3 MODEL-WISE RESULTS

This part presents the performance results of individual deep learning models implemented for diabetic retinopathy detection using the APTOS 2019 dataset. All models were trained and evaluated under similar experimental conditions to ensure fair comparison. The performance was assessed using validation accuracy and quadratic weighted kappa, which are suitable metrics for multi-class medical image classification.

5.3.1 ResNet50

The ResNet50 model demonstrated comparatively lower performance among the implemented architectures. The model achieved a validation accuracy of 60.25% and a quadratic weighted kappa score of 0.4598. While ResNet50 was able to learn basic retinal features, its performance was limited in distinguishing between adjacent severity levels of diabetic retinopathy. The confusion matrix analysis indicated higher misclassification between moderate and severe classes, suggesting insufficient feature discrimination for this dataset.

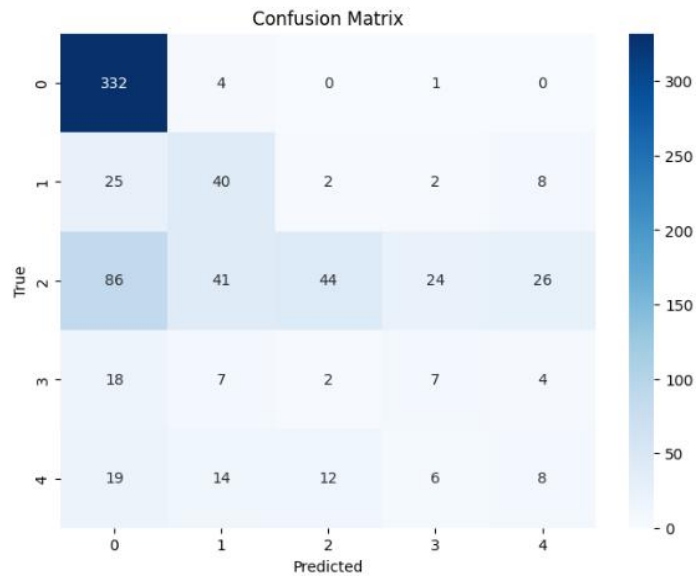


Fig. 5.1 Confusion Matrix for ResNet50 Model

5.3.2 EfficientNet-B0

EfficientNet-B0 showed noticeable improvement over ResNet50 in both accuracy and agreement metrics. The model achieved a validation accuracy of 70.49% with a score of 0.8373. The compound scaling approach of EfficientNet-B0 allowed better feature extraction while maintaining computational efficiency. The confusion matrix revealed improved classification performance across most diabetic retinopathy stages, particularly in separating normal and mild cases.

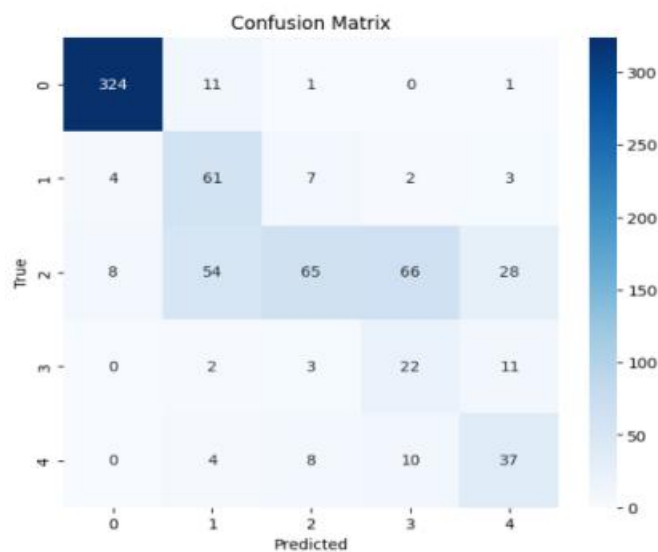


Fig. 5.2 Confusion Matrix for EfficientNet-B0 Model

5.3.3 DenseNet121

DenseNet121 delivered strong performance due to its dense connectivity and feature reuse capability. The model achieved a validation accuracy of 78.42% and score of 0.8571, indicating high agreement between predicted and actual labels. The dense feature propagation helped capture fine-grained retinal abnormalities, resulting in improved classification across moderate and severe stages. Training and validation curves showed stable convergence with minimal overfitting.

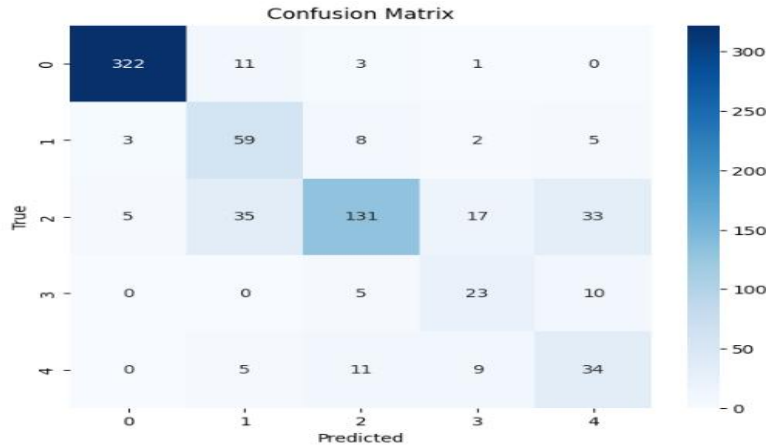


Fig. 5.3 Confusion Matrix for DenseNet121 Model

5.3.4 EfficientNet-B4

EfficientNet-B4 achieved the best overall performance among all implemented models. The model recorded a validation accuracy of 81.17% and a score of 0.8552. Its higher representational capacity enabled more accurate discrimination between different diabetic retinopathy severity levels. The model demonstrated consistent performance across all classes and showed superior generalization capability, making it the most effective architecture for the proposed system.

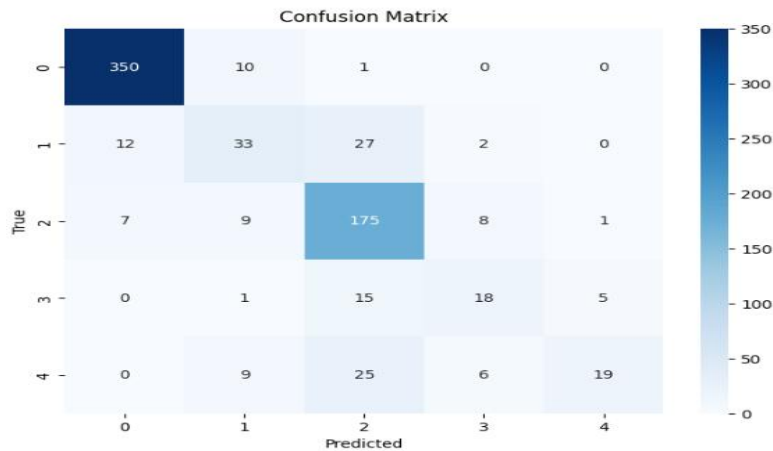


Fig. 5.4 Confusion Matrix for EfficientNet-B4 Model

5.4 COMPARISON OF MODELS

This part presents a comparative analysis of all deep learning models implemented for diabetic retinopathy detection using the APTOS 2019 dataset. The comparison is carried out to evaluate the relative performance of different architectures under the same experimental conditions. All models were trained using identical preprocessing techniques, dataset splits, and evaluation metrics to ensure a fair and unbiased comparison.

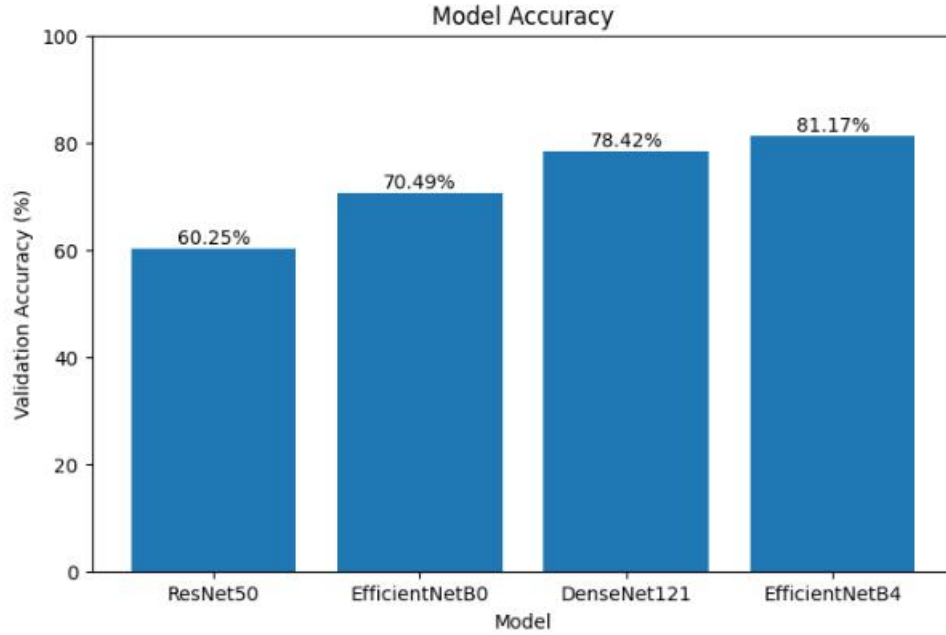


Fig. 5.5: Accuracy Comparison

ResNet50 exhibited the lowest performance among the evaluated models. Although it successfully learned basic retinal features, its ability to distinguish between closely related diabetic retinopathy severity levels was limited. This resulted in lower validation accuracy and Quadratic Weighted Kappa scores, indicating reduced agreement between predicted and actual labels. The deeper architecture without dense feature reuse may have affected its performance on this dataset.

EfficientNet-B0 demonstrated a significant improvement over ResNet50 by achieving better accuracy and consistency. Its compound scaling approach enabled efficient feature extraction with fewer parameters, resulting in improved classification performance. EfficientNet-B0 showed better generalization and reduced misclassification across most classes, making it a more suitable choice for resource-efficient implementations.

DenseNet121 delivered strong performance due to its dense connectivity, which promotes feature reuse and effective gradient flow. The model showed high accuracy and scores, particularly in distinguishing moderate and severe diabetic retinopathy stages. Its stable convergence and consistent predictions indicate its effectiveness in medical image classification tasks.

EfficientNet-B4 achieved the best overall performance among all implemented models. Its increased depth, width, and resolution allowed the model to capture fine-grained retinal features more effectively. The model demonstrated superior accuracy, strong agreement with ground truth labels, and consistent performance across all severity classes. As a result, EfficientNet-B4 was identified as the most effective architecture for the proposed diabetic retinopathy detection system.

Model Architecture	Validation Accuracy (%)	Quadratic Weighted Kappa (QWK)
ResNet50	60.25	0.4598
EfficientNetB0	70.49	0.8373
DenseNet121	78.42	0.8571
EfficientNetB4	81.17	0.8552

Table 5.1: Comparison of Model Performance

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

This project presented the design and implementation of an automated system for the detection and classification of diabetic retinopathy using deep learning techniques. By leveraging the APTOS 2019 retinal fundus image dataset and transfer learning-based convolutional neural network models, the system aimed to reduce manual dependency in retinal screening and improve diagnostic accuracy. The proposed approach focused on systematic data preprocessing, robust model training, and comprehensive performance evaluation.

Multiple deep learning architectures, including ResNet50, DenseNet121, EfficientNet-B0, and EfficientNet-B4, were implemented and evaluated under consistent experimental conditions. The comparative analysis demonstrated that advanced architectures with better feature representation capabilities achieved higher classification accuracy. Among the evaluated models, EfficientNet-B4 showed the best overall performance, indicating its effectiveness in capturing fine-grained retinal features associated with different stages of diabetic retinopathy.

The results obtained from this project confirm that deep learning-based approaches can serve as reliable decision-support tools for diabetic retinopathy screening. The system successfully addresses key limitations of traditional manual diagnosis, such as time consumption and inter-observer variability. Overall, the proposed system provides a scalable and efficient framework that can assist ophthalmologists in early detection and support large-scale retinal screening programs.

6.2 LIMITATIONS

Despite the successful implementation of the proposed diabetic retinopathy detection system, certain limitations were observed during the course of the project. One of the primary limitations is the dependency on the quality of retinal fundus images. Variations in illumination, focus, and image resolution can affect model performance, as deep learning models are sensitive to input image quality.

Finally, the proposed system functions as a decision-support tool and is not intended to replace professional medical diagnosis. Clinical validation and real-world deployment would require extensive testing, regulatory approval, and integration with healthcare systems, which were beyond the scope of this project.

6.3 FUTURE ENHANCEMENTS

The proposed diabetic retinopathy detection system can be further enhanced in several ways to improve accuracy, usability, and real-world applicability. One major enhancement involves training the system on larger and more diverse datasets collected from multiple sources. This would help improve model generalization and ensure robust performance across different imaging conditions and patient populations.

Advanced preprocessing and feature enhancement techniques can be incorporated to further improve detection of subtle retinal abnormalities. Future work may include the use of attention mechanisms and ensemble learning approaches to enhance feature representation and classification performance. Additionally, optimizing hyperparameters and exploring newer deep learning architectures could lead to further improvements in accuracy and reliability.

The system can also be extended to support real-time deployment through web-based or mobile applications. Integrating the model into clinical workflows would enable faster screening and assist ophthalmologists in early diagnosis. Furthermore, the system can be expanded to detect other retinal diseases such as glaucoma and age-related macular degeneration with minimal modifications.

Finally, future enhancements may focus on clinical validation and interpretability. Incorporating explainable AI techniques can help visualize important retinal regions influencing model predictions, thereby increasing trust and acceptance among medical professionals. With these improvements, the proposed system has the potential to become a valuable tool for large-scale retinal disease screening and healthcare support.

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